

1 Logik, Mengen, Zahlen

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2 Vectors and Matrices

2.1 Vectors

2.1.1 Linear Combination $\lambda \mathbf{v} + \mu \mathbf{w}$

$\mathbf{v}, \mathbf{w} \in \mathbb{R}^m, \lambda, \mu \in \mathbb{R}$
 $\sum_{i=1}^n \lambda_i \mathbf{v}_i$, e.g. $5 \begin{pmatrix} 2 \\ 3 \end{pmatrix} - 3 \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 10 \\ 15 \end{pmatrix} - \begin{pmatrix} 9 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 18 \end{pmatrix}$
Every vector in $\mathbb{R}^m$ can be written as $\sum_{i=1}^m u_i \mathbf{e}_i$.

- (i) **affine**: if $\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$
- (ii) **conic**: if $\lambda_j \geq 0$ for $j = 1, 2, \dots, n$
- (iii) **convex**: if both *affine* and *conic* combination

2.1.2 Lengths and Angles from Dot Products

Scalar (or dot, inner) product : $\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \dots + v_n w_n$.

Outer product $rank(A) = 1 \iff \exists$ non-zero vectors $\mathbf{v} \in \mathbb{R}^m, \mathbf{w} \in \mathbb{R}^n$, s.t. A is an outer product, i.e. $A = \mathbf{v}\mathbf{w}^\top$, thus $rank(\mathbf{v}\mathbf{w}^\top) = 1$.

Length (Euclidian Norm): $||\mathbf{v}|| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{\mathbf{v}^\top \mathbf{v}} = \sqrt{v_1^2 + \dots + v_n^2}$. $||\mathbf{v}||^2 := \mathbf{v}^\top \mathbf{v}$

Unit vector : $||\mathbf{u}|| = 1 = \frac{\mathbf{v}}{||\mathbf{v}||}$, for $\mathbf{v} \neq \mathbf{0}$.

Orthogonal (perpendicular) vectors: $\mathbf{v} \cdot \mathbf{w} = 0$.

Angle between vectors (Cosine Formula): $\cos \alpha = \frac{\mathbf{v} \cdot \mathbf{w}}{||\mathbf{v}|| ||\mathbf{w}||}$, for $\mathbf{v} \neq \mathbf{0}$.

Because $\cos \alpha \leq 1$, **Cauchy-Schwarz** inequality:

$$\underbrace{|\mathbf{v} \cdot \mathbf{w}|} \leq ||\mathbf{v}|| ||\mathbf{w}||$$
$$|\cos \alpha| ||\mathbf{v}|| ||\mathbf{w}||$$

Triangle inequality : $||\mathbf{v} + \mathbf{w}|| \leq ||\mathbf{v}|| + ||\mathbf{w}||$.

2.1.3 Linear (In)dependence of Vectors

$\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k$ are linearly **independent** if:

- (i) no vector is a linear combination of the previous ones. Or
- (ii) no vector is a linear combination of the other ones. Or
- (iii) there are no $c_1, c_2, \dots, c_k$ besides $0, 0, \dots, 0$ such that $c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + \dots + c_k \mathbf{w}_k = \mathbf{0}$.

All are equivalent: (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii).
Replace no by **some** for linear **dependence**.

A has **independent columns** if (iii) there is no $\mathbf{x}$ besides $\mathbf{0}$ such that $A\mathbf{x} = \mathbf{0}$.

2.1.4 Span

Span of vectors is the set of all linear combinations of those vectors.

2.1.5 Hyperplane through origin

Let $\mathbf{d} \in \mathbb{R}^m, \mathbf{d} \neq \mathbf{0}, H_{\mathbf{d}} = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{v} \cdot \mathbf{d} = 0\}$

2.2 Matrices Basics

$A \in \mathbb{R}^{m \times n}$: a **matrix** A with m rows, n columns. *Zeilen matrix, Spalten matrix*
 $A = [a_{ij}]_{i=1, j=1}^m, n$

Trace : Sum of the diagonal entries.

rank(A) = # independent columns (rows).

2.2.1 Distributivity and associativity

$A(B+C) = AB+AC, (B+C)D = BD+CD$. And $(AB)C = A(BC) = ABC, (AB)(CD) = A((BC)D) = \dots = ABCD$. Matrix multiplication isn't commutative.

2.3 The Transpose of A

$$\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}^\top = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix} \quad \left| \quad \begin{array}{l} \text{Row } i \text{ of } A = \text{column } i \text{ of } A^\top, \\ \text{column } j \text{ of } A = \text{row } j \text{ of } A^\top. \end{array} \right.$$

Product: $(AB)^\top = B^\top A^\top$.

More matrices: $(ABC)^\top = C^\top B^\top A^\top$.

Inverse: $(A^{-1})^\top = (A^\top)^{-1}$.

Proof: $(A^{-1})^\top A^\top = (AA^{-1})^\top = I^\top = I$.

Permutation: $P^\top = P^{-1}$.

Proof: $\mathbf{p}_i \cdot \mathbf{p}_j = I_{ij} \Leftrightarrow PP^\top = I$.

2.4 Inverse Matrices

$M \in \mathbb{R}^{n \times n}$ (square!) is invertible if there is a matrix $M^{-1} \in \mathbb{R}^{n \times n}$ (the inverse of M) such that $MM^{-1} = M^{-1}M = I$.
There can only be **one inverse**: If $MX = YM = I$, then $X = IX = (YM)X = Y(MX) = YI = Y$.

$$\mathbb{R}^{1 \times 1}: [x]^{-1} = \left[\frac{1}{x}\right] \text{ (if } x \neq 0)$$

$$\mathbb{R}^{2 \times 2}: \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \text{ if } (ad-bc \neq 0)$$

$$\mathbb{R}^{n \times n}: [M|I] \rightarrow [I|M^{-1}] \text{ using Gauss-Jordan elimination (3.2.1)}$$

2.4.1 The Inverse Theorem

$A \in \mathbb{R}^{n \times n}$ is invertible
 $\Leftrightarrow \forall \mathbf{b} \in \mathbb{R}^n, A\mathbf{x} = \mathbf{0}$ has a unique solution $\mathbf{x}$
 $\Leftrightarrow$ The columns of A are independent.

For any $A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times n}$, if $AB = I$, then $BA = I$.

Proof: $AB = I \Leftrightarrow B = A^{-1} \Leftrightarrow BA = A^{-1}A = I$.

2.4.2 Properties of the inverse

If $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times n}$ are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$. *Proof*: $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = I$.

More matrices: $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

$$(A^{-1})^{-1} = A$$

$$(A^\top)^{-1} = (A^{-1})^\top$$

2.5 Special Matrices

- **Identity Matrix** ($a_{ii} = 1$ for all i): I
- **Diagonal Matrix** ($a_{ij} = 0$ for all $i \neq j$)
- **Upper triangular matrix** ($a_{ij} = 0$ for all $i > j$) : U
- **Lower triangular matrix** ($a_{ij} = 0$ for all $i < j$) : L
- **Symmetric matrix** ($a_{ij} = a_{ji}$ for all i, j) : $A = A^\top$
- **Skew-symmetric matrix** ($a_{ij} = -a_{ji}$ for all i, j) : $A = -A^\top$
- **Nilpotent matrix** $A^k = 0$ for some $k \in \mathbb{N}$

2.5.1 Symmetric Matrices

A matrix $S_{n \times n}$ satisfying $S = S^\top$ is symmetric.

If S is symmetric, then S^{-1} is also symmetric (if it exists).

Proof: $S^{-1} = (S^\top)^{-1} = (S^{-1})^\top$.

$AA^\top$ (and $A^\top A$) is symmetric.

Proof: $AA^\top = (A^\top)^\top A^\top = (AA^\top)^\top$.

2.6 CR-Decomposition $A = CR$

$C \in \mathbb{R}^{m \times r}$: the independent columns.

$R \in \mathbb{R}^{r \times n}$: how to combine them to get all columns.
 $r = rank(A)$

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

Interpretation: $\mathbf{v}_1 = 1\mathbf{e}_1 + 0\mathbf{e}_2, \dots, \mathbf{v}_4 = 3\mathbf{e}_1 + -2\mathbf{e}_2$.

Computation: Gauss-Jordan elimination (3.2.1). Get R from RREF. C is the columns from A where there is a pivot in R .

3 Trigonometry

